QP CODE: 25900054

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Course (MDC)- MG2MDCCFS102 - INTRODUCTION TO COMPUTER NETWORKS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 35**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Short Answer Type Questions

Answer **any 10** questions

Each question carries **1** mark

1. Error-reporting and diagnostic protocol, which supports the functionality of the IP protocol within the network layer - 1 mark. [U]/ [CO 1]
2. The easiest network topology to troubleshoot is the Star Topology**.** - 1 mark. [An]/ [CO 1, CO 2]

3. The routers are responsible for forwarding data packets between different network segments and determining the best path for data to travel across a WAN - 1 mark.[U]/[CO1]

4. The IPv4 header is 20 to 60 bytes long, as it has optional fields. It includes fields like checksum, identification, flags, and fragment offset, which help in error-checking and fragmentation of data packets.

The IPv6 header is fixed at 40 bytes, with no optional fields , making it more streamlined and efficient. - 1 mark. [U]/ [CO 1]

5. Primary Server: Holds the original and authoritative of DNS records , makes changes to DNS records, Primary Server is the "master" server for DNS data.

Secondary Server: Provides fault tolerance and redundancy, ensuring DNS availability even if the primary server fails. - 1 mark. [U]/ [CO 2]

6. The DATA command in SMTP is a key component of the message transfer process. It is used to indicate the beginning of the email message content that the sender wants to deliver.

- 1 mark. [U]/[CO2]

7. Backpressure helps manage data flow to prevent network congestion and ensure smooth communication between devices in a network**.** - 1 mark. [An]/ [CO 2]

8. The goal of security architecture is to create a secure, robust, and resilient environment for data transmission and communication, defending against potential threats and attacks.

- 1 mark.[U]/[CO1]

9. Types : security services and security mechanisms that can be implemented across layers to ensure the confidentiality, integrity, authenticity, and availability of data. - 1 mark. [U]/ [CO 1]

10**.** Two types of access control**:-** Discretionary Access Control (DAC) and Mandatory Access Control (MAC)**.** - 1 mark. [U]/ [CO 1]

11. Deadlock is a condition where a set of processes are in a state of perpetual waiting, unable to make any progress **.** - 1 mark. [U]/ [CO 4]

12. Digital evidence can be categorized into volatile and non-volatile types, and they differ significantly in terms of how they are collected, preserved, and analyzed. - 1 mark.[An]/[CO 4]

(1 x 10 = 10)

Part B

Short Essay Type Questions

Answer **any 3** questions

Each question carries **5** marks

13. List significance of network protocols - 2 marks

Explanation with examples - 3 marks

14. Checksum operation in error detection - 1 mark

Sender's action (Calculation of checksum and transmission) with example - 2 mark

Receiver's action (Checksum Validation and Error Detection) with example - 2 mark

15. Hamming Code for Error Correction - 1 mark

Explain with example for Hamming Code for Error Correction - 4

16. Active Security and Passive Security Attacks - 1 mark

Characteristics of active and passive security attacks - 2 mark

Examples of active and passive security attacks - 2 mark

(3 × 5 = 15)

Part C

Essay Type Questions

Answer **any 1** question

Each question carries **10** marks

17.Different connecting devices (repeater**,** hub**,** bridge**,** switches**,** router**,** gateway etc) - 1 mark

Explain each connecting devices ,write functions - 4 mark

Uses, advantages and examples - 5 mark

18. Go-Back-N ARQ - 2 mark

Piggybacking Concept - 2 mark

Piggybacking Works in G-Back-N ARQ - 3 mark

Example of Piggybacking in G-Back-N ARQ - 3 mark

(10 × 1 = 10)